Table 1.3 lists some of the derived quantities and their units.

Table 1.3 Derived Quantities and their Units

Physical quantity	Expression	Unit
Plane angle	arc / radius	rad
Solid angle	surface area/radius^2	sr
Area	length × breadth	m^2
Volume	area × height	m^3
Velocity	displacement / time	$m\ s^{-1}$
Acceleration	velocity / time	$m\ s^{-2}$
Angular velocity	angular displacement / time	$rad\ s^{-1}$
Angular acceleration	angular velocity / time	$rad\ s^{-2}$
Density	mass / volume	$kg\ m^{-3}$
Linear momentum	mass × velocity	$kg\ m\ s^{-1}$
Moment of inertia	mass × (distance)^2	$kg\ m^2$
Force	mass × acceleration	$kg\ m\ s^{-2}$ or N
Pressure	force / area	$N\ m^{-2}$ or Pa
Energy (work)	force × distance	N m or J
Power	Work / time	$J\ s^{-1}$ or watt (W)
Impulse	force × time	N s
Surface tension	force / length	$N\ m^{-1}$
Moment of force (torque)	force × distance	N m
Electric charge	current × time	A s or C
Current density	current / area	$A\ m^{-2}$
Magnetic induction	force / (current × length)	$N\ A^{-1}\ m^{-1}$ or tesla
Force constant	force / displacement	$N\ m^{-1}$
Plank's constant	energy of photon / frequency	J s
Specific heat (S)	heat energy / (mass × temperature)	$J\ kg^{-1}\ K^{-1}$
Boltzmann constant (k)	energy/temperature	$J\ K^{-1}$

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